Kiran Muppana

Tallahassee, FL | +1 (850) 943-4453| kiranmuppana36@gmail.com | LinkedIn | Github

Experience

Software Engineer 2, Florida State University Oct 2024 – Present

- Engineered a scalable real-time energy management system using Java, SpringBoot, Kafka and Kubernetes.
- Led critical React TypeScript **performance** optimization, reducing the production bundle size by 70% and initial route size by 94% using React Dynamic Imports and code splitting to boost LCP and overall SEO.
- Configured runtime **caching** rules with HTTPS Cache-Control headers and React Query client-side caching, cutting average server costs 60% and DB queries 70%
- Orchestrated horizontal sharding strategy for **PostgreSQL** clusters, improving write performance by 2x.
- Developed hierarchical knowledge graph search algorithm enabling **LLM** vector semantic searches via **MongoDB** Vector DB, improving query relevance and retrieval accuracy for complex relational data.
- Configured **Jenkins** CI/CD builds and Helm charts, handling over **5M** daily API requests with 99.9% availability.

Software Engineer, Florida State University Dec 2022 – Oct 2024

- Developed reactive, non-blocking APIs **Java Spring Webflux**, enabling high-throughput async service-to-service communication and reducing p95/p99 latency by 65%.
- Created autoscaling mechanisms on **AWS ECS** and **Fargate**, optimizing compute cost by 25%.
- Collaborated across 4 teams to standardize API contracts and enforce schema consistency with **OpenAPI 3.0**.
- Implemented Prometheus/Grafana **observability** with namespace isolation, reducing MTTR to under 15 mins.
- Deployed **Redis** for distributed caching and optimized key eviction policies, improving app responsiveness by 45%.
- Architected an AI workflow analyzing 150k documents to auto-generate work orders, eliminating manual steps and saving 25+ hours weekly.

Software Engineer, Amdocs Jul 2021 – Jul 2022

- Devloped RESTful **AIOps** microservice using **Java SpringBoot**, implementing layered traffic management (soft/hard rate limiting) and advanced failure protection via Circuit Breaking and Bulkhead isolation.
- Integrated react **virtualization** and **lazy loading** for infinite scroll, rendering only visible items to cut DOM nodes 80% and boost scroll performance on low-end devices.
- Optimized **GraphQL** queries and DB indexes, reducing report generation time from 3 minutes to under 30 secs.
- Automated secure authentication using OAuth, JWT, and React Routes, ensuring seamless role-based navigation.
- Worked closely with QA to adopt **Test-Driven Development (TDD)** practices across the team.
- Built automated deployment pipelines with **Jenkins**, **Terraform** and integrated quality gates using **SonarQube**.
- Worked in Agile teams with regular code reviews, sprint planning, retro and cross-functional collaboration.

Technical Skills

Programming Languages: Java, Go, Python, JavaScript, SQL, Shell
Backend Frameworks: Spring Boot, WebFlux, Multi-Threading, gRPC, Gin, Kafka Streams
Frontend: React, Next.js, TypeScript, Tailwind CSS, LightHouse, Chrome Dev Tools
Cloud/DevOps: AWS (ECS, Lambda, Kinesis), Kubernetes, Terraform, Docker, Jenkins, GitHub Actions
Databases/Observability: PostgreSQL, Oracle DB, MongoDB, Redis, Snowflake, Prometheus, Grafana
AI/ML: LangGraph, LangChain, LlamaIndex, PyTorch, TensorFlow, LLM, RAG

Education

Florida State University, Tallahassee, FL – Master's in Computer Science
National Institute of Technology, Calicut, India – Bachelor's in Electronics & Communication

Certifications & Publications

Certifications: Oracle Certified Generative AI Professional | Oracle Certified Data Science Professional
Publication: *Transformer models for enhancing AttnGAN based text to image generation* – Elsevier